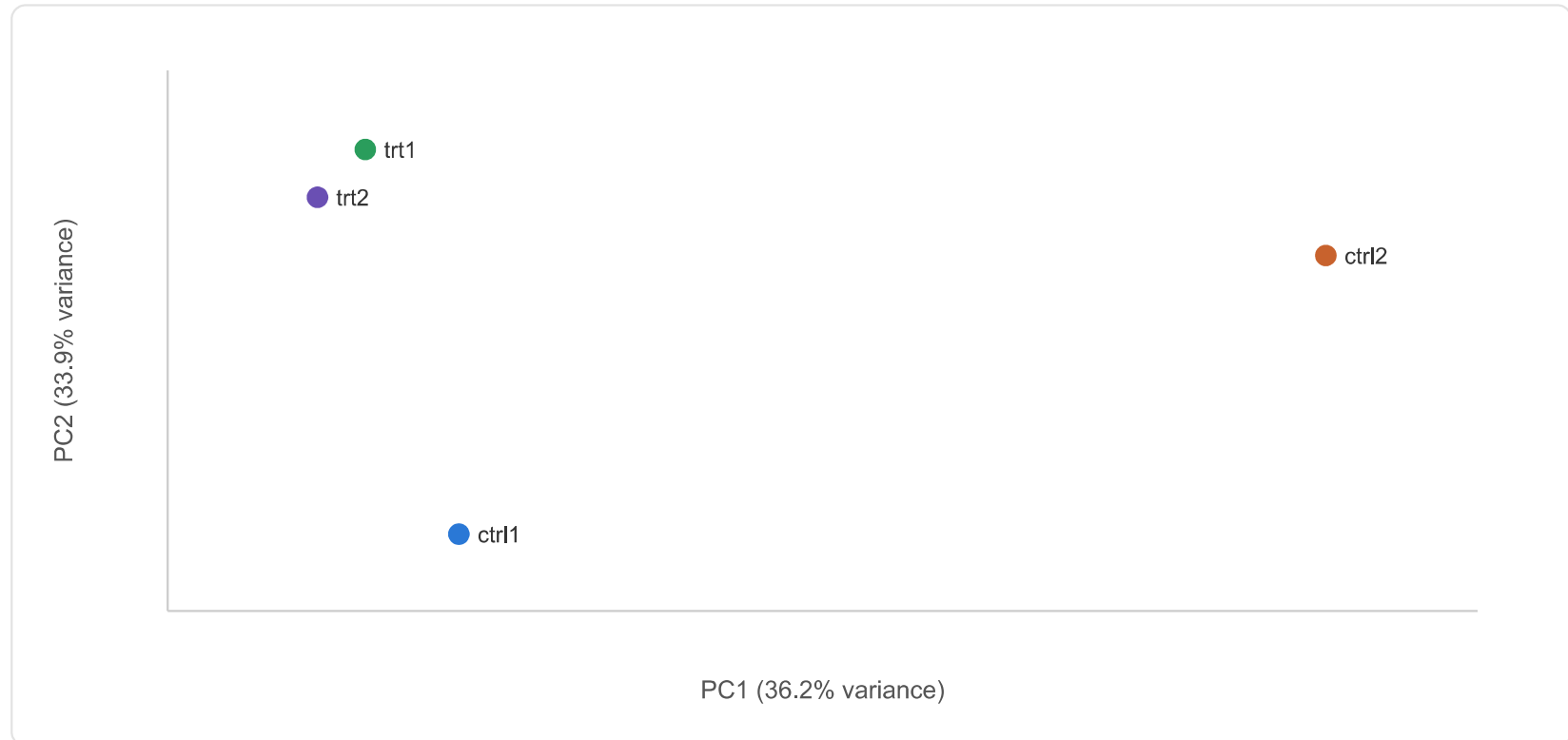


Multi-sample comparison

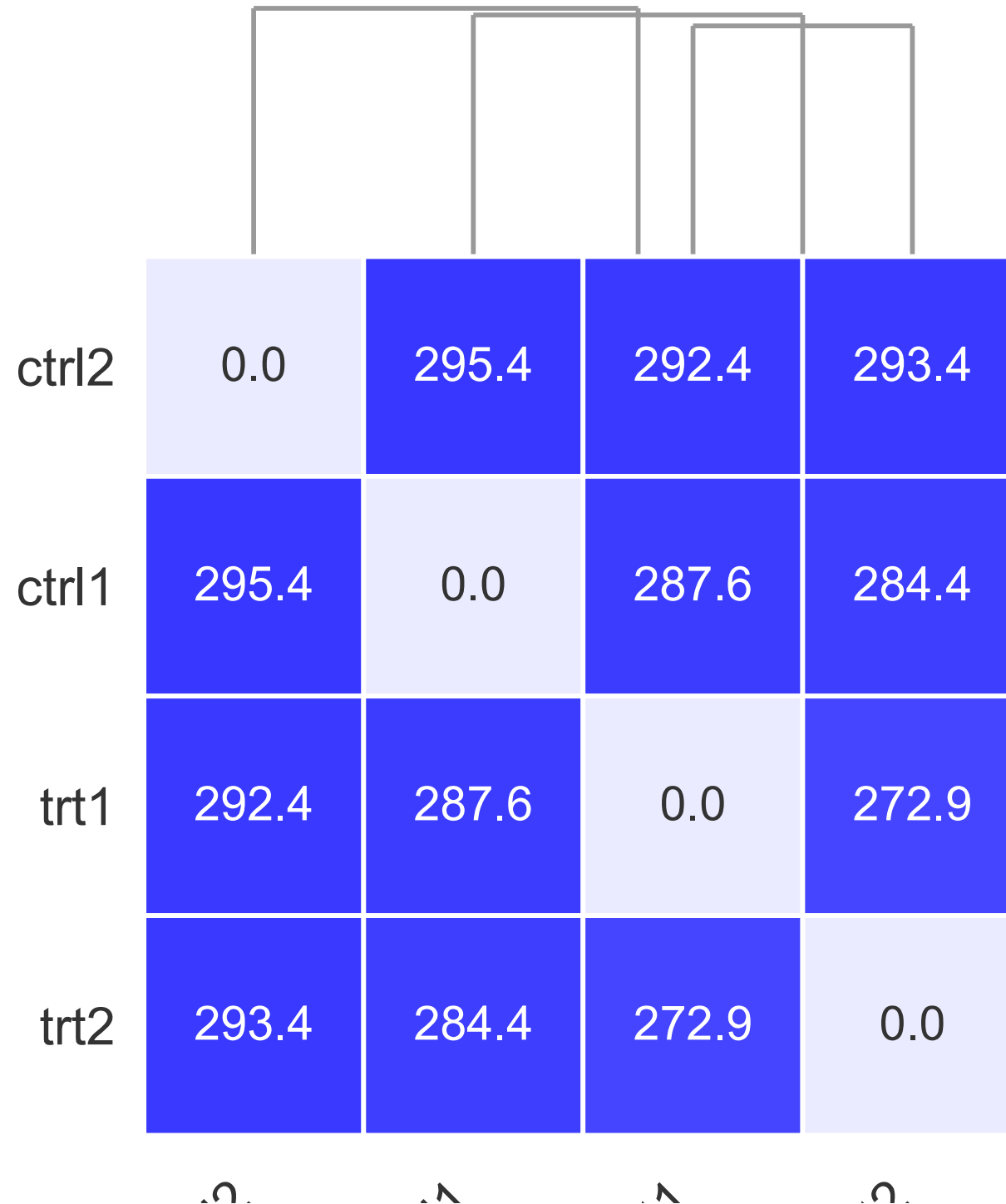
4 samples · top 500 most-variable genes (of 7127)

Sample similarity (PCA)



Each point is one sample, projected onto its first two principal components computed from $\log_2(\text{size-factor-normalized count} + 1)$ over the top 500 most-variable genes (exact eigendecomposition of the sample-similarity matrix, the standard approach when there are far more genes than samples -- not an approximation). Samples that cluster together have more similar overall expression profiles.

Sample-distance heatmap



ctrl

ctrl

trt

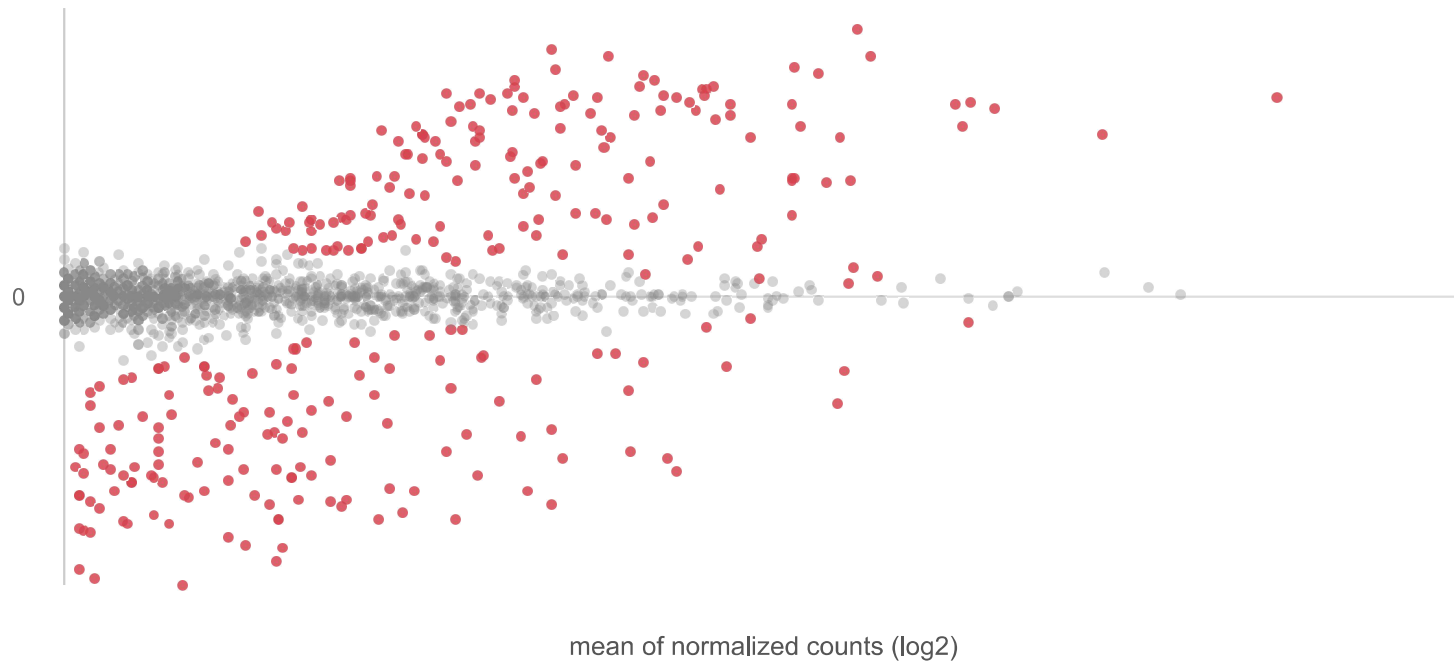
trt

Pairwise Euclidean distance between samples in the same top-500-variable-gene VST-transformed space used for the PCA above (darker = more similar). Rows/columns are ordered by average-linkage hierarchical clustering, shown as the dendrogram on top -- samples that merge low are more similar. This uses the same top-variable-gene subset as the PCA for consistency; DESeq2's own sample-distance heatmap conventionally uses the full transformed matrix instead.

Differential expression: treated vs control

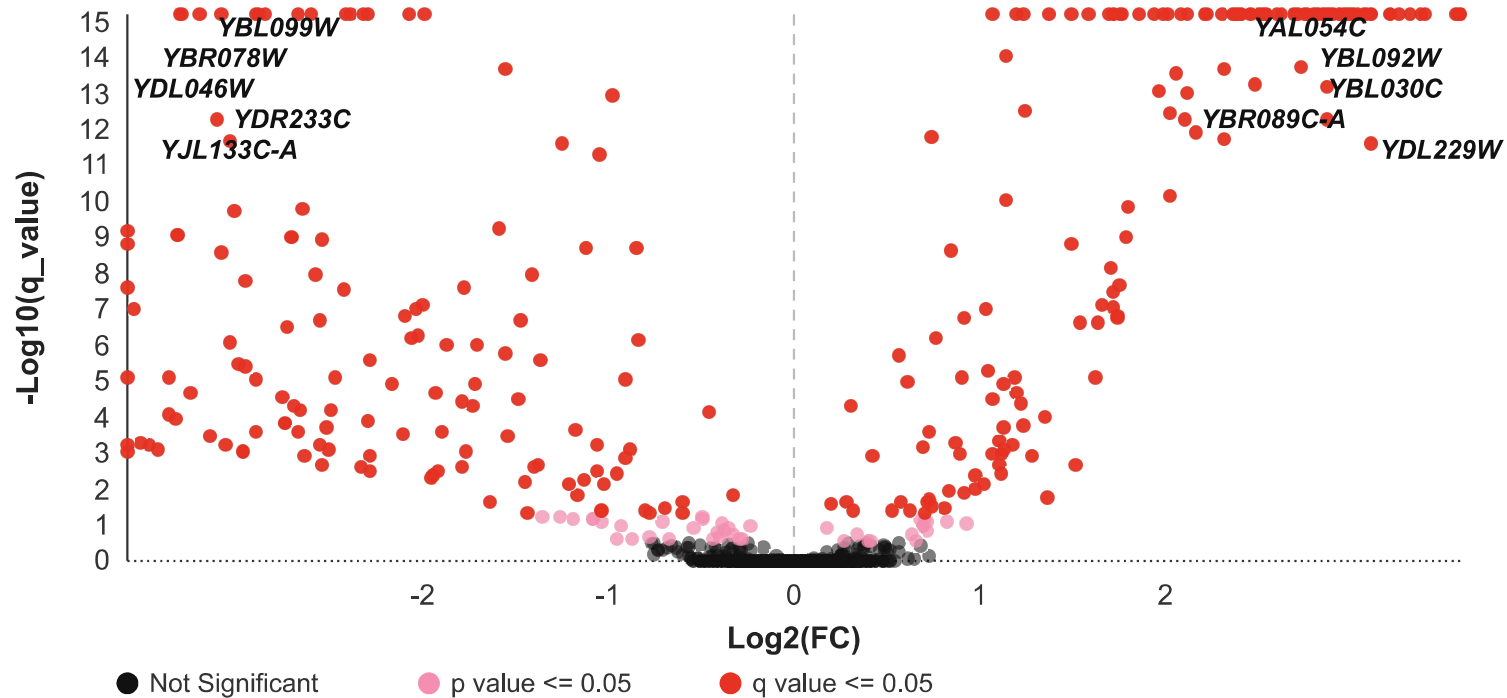
1784 genes tested after independent filtering (of 7127 total) · 303 significant at FDR < 0.1 · positive log2FoldChange means higher in "treated"

Note on the "log2FC (shrunk)" column: the estimated LFC prior variance collapsed to essentially zero for this comparison. That happens (correctly, not as a bug) when the overwhelming majority of tested genes show no real difference between conditions -- with so little genuine between-gene effect variance to estimate, DESeq2's classic "normal" shrinkage pulls even confident true positives most of the way to zero. This is a known, documented limitation of the "normal" method specifically (it's the reason DESeq2 itself switched its default to apeglm, which isn't implemented here -- see the module notes in the source). **Significance calling (p-value/padj) is unaffected** -- that's based on the unshrunk Wald test throughout. For effect size, trust the "log2FC (MLE)" column and the MA/volcano plots (both use the unshrunk value) over the shrunk column here.



Each point is one gene, y-axis is the unshrunk log2FoldChange (MLE). Red = significant at FDR < 0.1. Genes dropped by independent filtering are not shown. The results table below also reports the empirical-Bayes-shrunk value in a separate column -- see the note above the table for when to trust it.

Volcano Plot



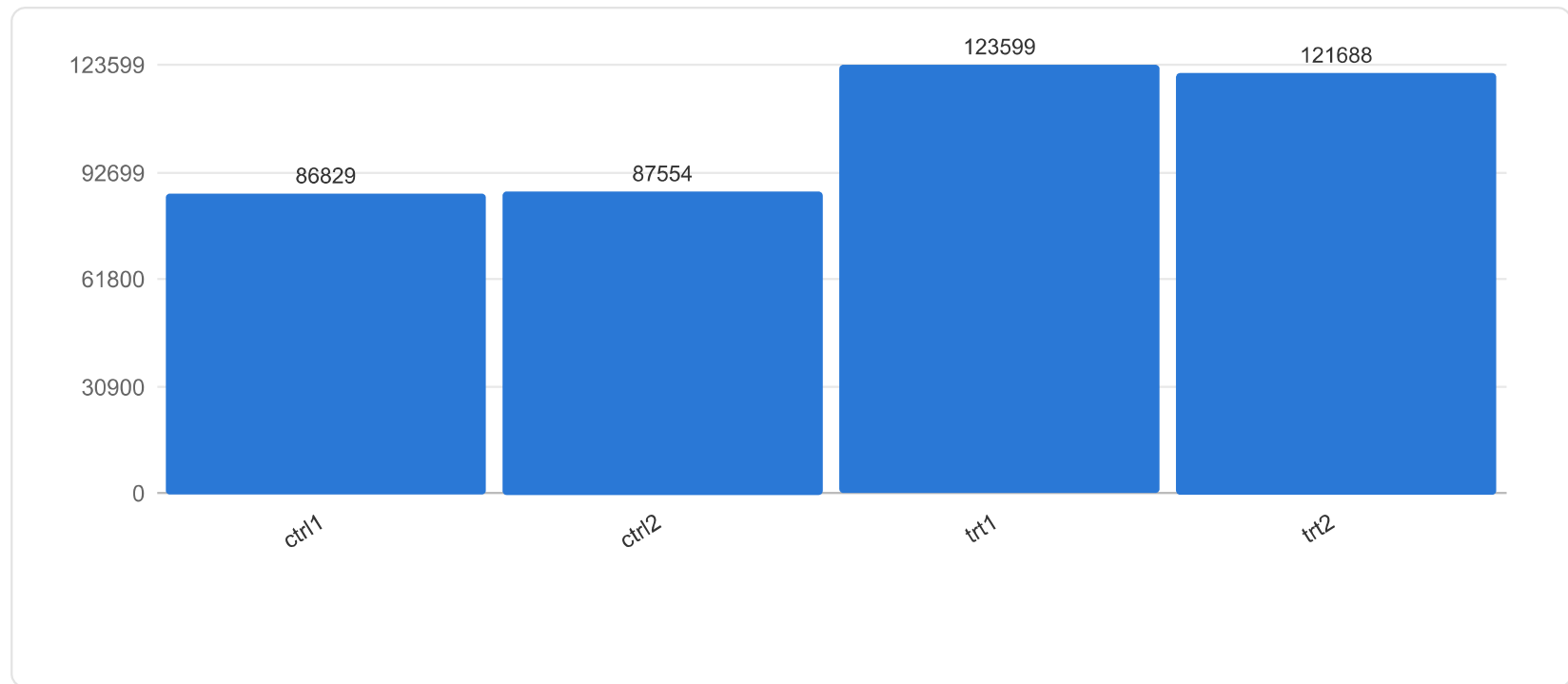
Each point is one gene passing independent filtering; x = log2FoldChange (unshrunk MLE), y = -log10(padj/q-value). Note the threshold here is $q \leq 0.05$, stricter than the $\text{padj} < 0.1$ this report's results table and other plots use as "significant" -- matching the conventional 0.05 threshold this plot style is usually drawn with, not a change to this report's own significance calls elsewhere. "p value <= 0.05" (pink) is uncorrected for multiple testing and shown for reference only, not as a second significance tier -- the same distinction DESeq2's own documentation draws between results()'s pvalue and padj columns. The x-axis range is bounded by the 99th percentile of $|\log_2\text{FC}|$ among adequately-expressed genes ($\text{baseMean} \geq 10$), not the single largest value over every gene -- a small number of very-low-count genes can otherwise produce huge but statistically unsupported fold-change estimates that would dominate the whole plot's scale. 0 non-significant gene(s) with an off-scale estimate under that bound are omitted from this plot rather than misleadingly drawn at the edge (never applies to $q \leq 0.05$ genes, which are always shown, clamped to the edge if needed) -- see gene_counts.tsv/the full results table for their actual values. Labeled genes: the 5 most significant up- and down-regulated genes by q-value.

Top 30 genes by adjusted p-value

gene_id	baseMean	log2FC (MLE)	log2FC (shrunk)	lfcSE	stat	pvalue	padj	dispersion	Cook's flag
YAL054C	1371.8	2.398	0.008	0.257	9.35	0.00e+00	0.00e+00	0.030	
YBL099W	99.5	-2.384	-0.007	0.267	-8.93	0.00e+00	0.00e+00	0.016	
YBL092W	191.2	2.750	0.009	0.247	11.14	0.00e+00	0.00e+00	0.018	
YBL030C	816.0	2.791	0.074	0.088	31.85	0.00e+00	0.00e+00	0.001	

YBR078W	66.0	-2.663	-0.008	0.259	-10.29	0.00e+00	0.00e+00	0.000
YBR089C-A	54.8	2.118	0.007	0.250	8.48	0.00e+00	0.00e+00	0.000
YDL229W	197.0	3.067	0.023	0.167	18.42	0.00e+00	0.00e+00	0.000
YDL181W	418.5	3.931	0.026	0.178	22.14	0.00e+00	0.00e+00	0.005
YDL125C	185.5	2.875	0.022	0.163	17.64	0.00e+00	0.00e+00	0.000
YDL046W	42.8	-2.837	-0.005	0.336	-8.44	0.00e+00	0.00e+00	0.000
YDR035W	63.5	2.847	0.008	0.277	10.29	0.00e+00	0.00e+00	0.000
YDR224C	152.2	2.012	0.016	0.162	12.41	0.00e+00	0.00e+00	0.002
YDR233C	56.5	-2.307	-0.007	0.257	-8.99	0.00e+00	0.00e+00	0.000
YDR384C	407.8	1.714	0.036	0.099	17.37	0.00e+00	0.00e+00	0.001
YDR450W	79.2	1.767	0.010	0.193	9.13	0.00e+00	0.00e+00	0.000
YEL058W	78.8	3.095	0.009	0.265	11.66	0.00e+00	0.00e+00	0.000
YEL007W	94.5	3.644	0.009	0.283	12.86	0.00e+00	0.00e+00	0.000
YER009W	155.8	3.204	0.018	0.194	16.47	0.00e+00	0.00e+00	0.000
YER044C	126.8	2.374	0.016	0.174	13.64	0.00e+00	0.00e+00	0.000
YER124C	66.2	3.025	0.005	0.350	8.64	0.00e+00	0.00e+00	0.020
YER165W	209.8	2.611	0.018	0.175	14.92	0.00e+00	0.00e+00	0.005
YFL045C	121.8	2.201	0.009	0.220	9.99	0.00e+00	0.00e+00	0.009
YFL037W	77.2	2.083	0.010	0.209	9.98	0.00e+00	0.00e+00	0.000
YFL016C	79.0	3.201	0.009	0.273	11.73	0.00e+00	0.00e+00	0.000
YFR033C	142.2	1.077	0.013	0.129	8.32	0.00e+00	0.00e+00	0.000
YFR050C	201.5	3.066	0.023	0.165	18.62	0.00e+00	0.00e+00	0.000
YGL205W	695.0	2.539	0.059	0.094	27.12	0.00e+00	0.00e+00	0.001
YGL191W	96.2	1.492	0.011	0.167	8.92	0.00e+00	0.00e+00	0.000
YGL187C	163.2	1.372	0.018	0.126	10.89	0.00e+00	0.00e+00	0.000
YGR106C	124.8	3.560	0.011	0.262	13.60	0.00e+00	0.00e+00	0.005

Library size



Total EM-assigned effective counts per sample. Analogous to the "M Assigned" column in MultiQC's General Statistics table.

What this uses vs. real DESeq2

Size factors: DESeq2's real median-of-ratios method. PCA/heatmap: DESeq2's real closed-form variance-stabilizing transform, fit blind to the experimental design (same as `vst()` with `blind=TRUE`). Dispersion: Cox-Reid adjusted profile likelihood (the real per-gene MLE DESeq2 uses), with a Gamma-GLM-fitted trend and empirical-Bayes shrinkage using an estimated prior variance (not a fixed weight). LFC shrinkage: DESeq2's classic "normal" method (normal-normal conjugate posterior). Cook's distance: flagged against the exact $F(2, n-2)$ quantile DESeq2 uses. PCA itself is exact eigendecomposition, not approximated.